# Répétitoires : Vignettes

## AVC ischémiques
## Hémorragies intracrâniennes
## Malformation cerebrovasculaires

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# Vignette 1

A 56 year old hypertensive woman presents with acute left hemiparesis, and a stroke is suspected. Imaging will be performed shortly.

Which statement about the mechanism/etiology of stroke is <u>wrong</u> ?

A) Hypertension is a risk factors for both ischemic and haemorrhagic stroke
B) Clinically, ischemic cannot be differentiated reliably from haemorrhagic stroke
C) Work-up of ischemic stroke should include imaging of the cervical arteries
D) For long-term prevention of ischemic stroke without a cardiac source, antiplatelets or anticoagulants can be given

# Facteurs de risque principaux de l'AVC

| | AVC/AIT ischémique | Hémorragiques |
|---|---|---|
| Communs | Hypertension | Hypertension |
| | Tabagisme | Tabagisme |
| | Régime (y.i. sodium, sucre, ...) | Régime (y.i. sodium, sucre, ...) |
| | Obésité | Obésité |
| | Facteurs psychosociaux | Facteurs psychosociaux |
| Spécifiques | Diabète / prédiabète | Surconsommation d'alcool |
| | Dyslipidémie | Coagulopathies |
| | Sédentarisme | |
| | Maladies cardiaques (FA etc.) | |
| Non-modif. | Age | Age |
| | Génétique | Génétique |

INTERSTROKE / O'Donnell Lancet 2010 and 2016

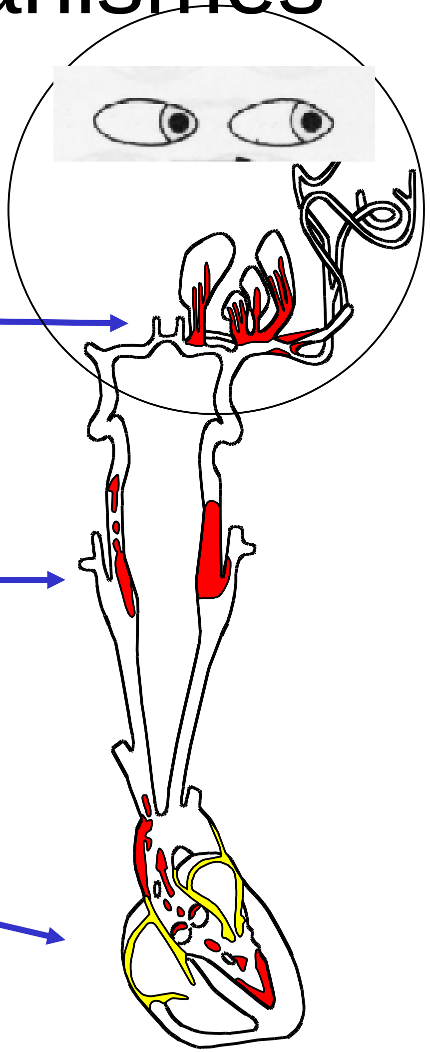
# AVC / AIT ischémique: mécanismes

« The big three » :

1) Micro-angiopathique
(« AVC lacunaire »)

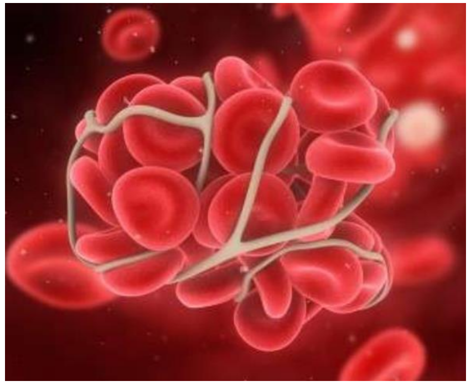
2) Athéromatose des artères extra > intracérébrales > crosse aortique

3) Cardio-embolique



Mumenthaler, Mattle
Thieme 2002

# Résumé : Quels antithrombotiques après AVC et AIT ?



- Anticoagulation si source cardiaque à haut risque (FA, valves méc, ....)
- Antiplaquettaires: pour tous les autres AVC, i.e.
  AVC sur athérosclérose/microangiopathie prouvée ou présumée, et d'origine indéterminée

Antiplaquettaire chronique préféré : Clopidogrel 75 mg/jour [1]

Patient à très haut risque :

- *Phase aiguë* (2 semaines) : Clopidogrel & Aspirine si sténose ou plaques athéromateuses instables symptomatiques [2]

1. Meta-analyse: Hankey Stroke 2000; 2. Meta-analyse Hao BMJ 2019; 3. COMPASS NEJM 2017

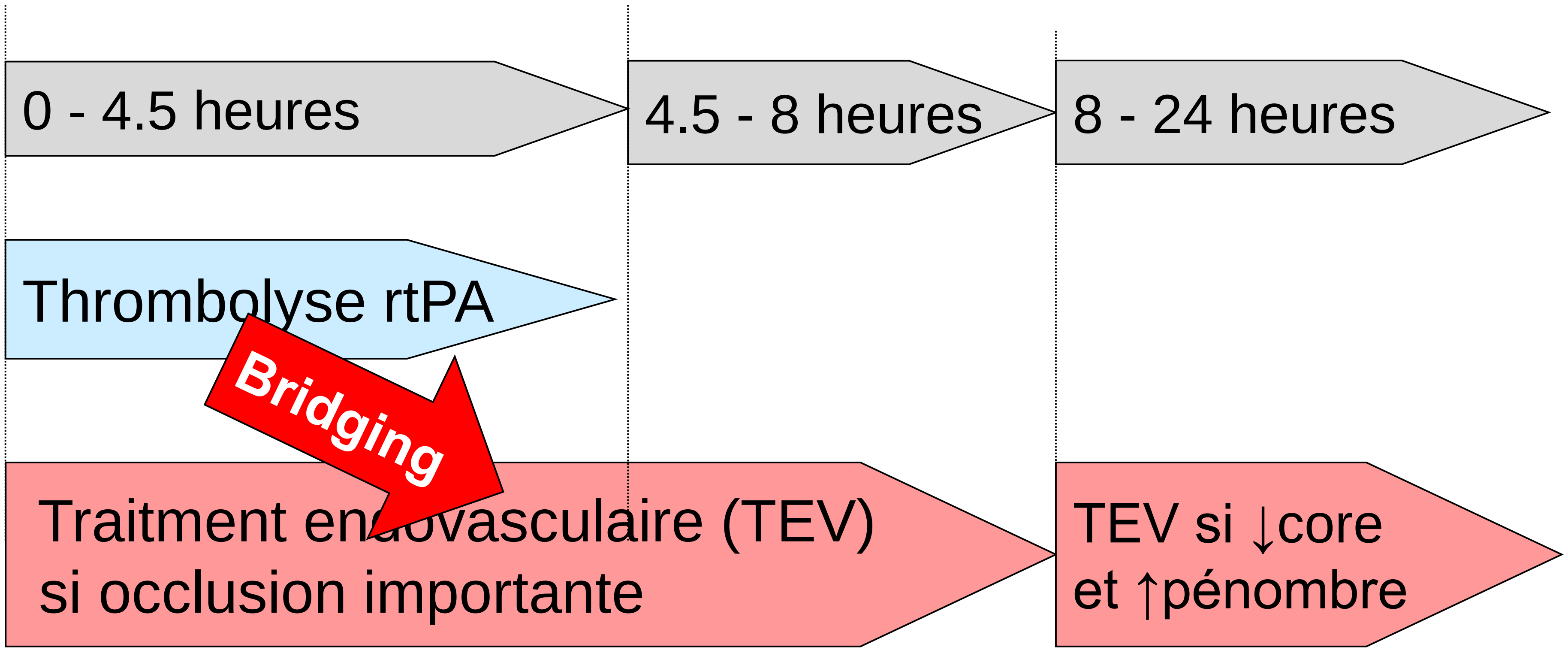
# Vignette 2

A 68 year old man has acute aphasia and right hemiparesis since one hour, and ischemic stroke is suspected.

Which statement is correct ?

A) The ambulance should bring the patient immediately to the nearest hospital for evaluation
B) The ambulance drivers should give an initial aspirine dose
C) The usual time window for IV thrombolysis (IVT) is 8 hours
D) If a proximal intracranial occlusion is found, endovascular treatment should be performed immediately after IVT

# Revascularisation aigu
## recommandée pour l'AVC ischémique en Suisse



# Vignette 3

A 55 year old hypertensive man, smoker, is found to have a TIA from a 90% carotid stenosis.

Regarding invasive secondary prevention, which statement is <u>correct</u> ?

A) Symptomatic carotid stenosis can treated by endarterectomy or by stent, according to patient profile

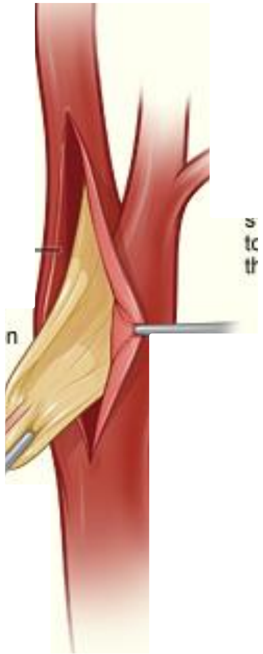
B) The carotid intervention should be performed the earliest at 2-4 weeks

C) Despite a normal echo, he should have long-term implantable cardiac monitoring to detect atrial fibrillation

D) This patient’s patent foramen ovale should also be closed

# Sténose carotidienne

## symptomatique et endartérectomie



- Endartérectomie pour sténose ≥ 70%
  - Ischémie récente
  - Espérance de vie > 2-3 ans
  - Chirurgien avec complications < 6%
- Endartérectomie pour sténose 50-69%
  - Idem, plaques instables

→ Opérer dans les 3-7 jours

# FOP
## Recommandations

- Bilan détaillé de l'AVC par experts cérébrovasculaires
- Estimer la probabilité que le FOP soit responsable de l'AVC (score « RoPE »)
- Offrir la fermeture après consentement informé si score «RoPE» élevé (≥ 6) = patient jeun, peu de factors de rique

RoPE score (Risk of Paradoxical Embolus), Kent Neurology 2013
Meta-analyse: Ntaios & Michel et al, Stroke 2018

# Vignette 4

A 62 year old man suffers acute right hemiparesis, with the acute CT showing a left deep hemispheric haemorrhage.

Which etiology is most likely ?

A) Cerebral vascular malformation
B) Cerebral amyloid angiopathy
C) Arterial hypertension
D) Cerebral vasculitis
E) Cerebral sino-venous thrombosis

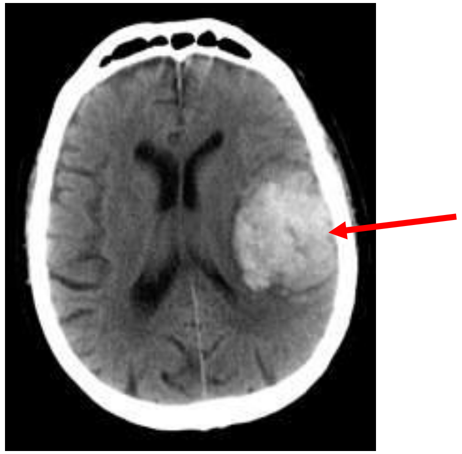
# HIC: Localisation → Etiologie

- Profonde et /ou intraventriculaire
  - ➢ Hypertension

IPP51360

IPP2233932

- Lobaire (superficielle)
  - ➢ <u>Toutes</u> les étiologies possible
  - ➢ Si postérieure : surtout angiopathie amyloïde
  - ➢ Si vertex ou temporale: surtout thrombose veineuse



IPP580717

# Vignette 5

A 46 year old woman suffers acute headache and decreased level of consciousness. Exam shows meningism but no focal signs. An acute CT shows subarachnoid hemorrhage (SAH) in the base of the brain.

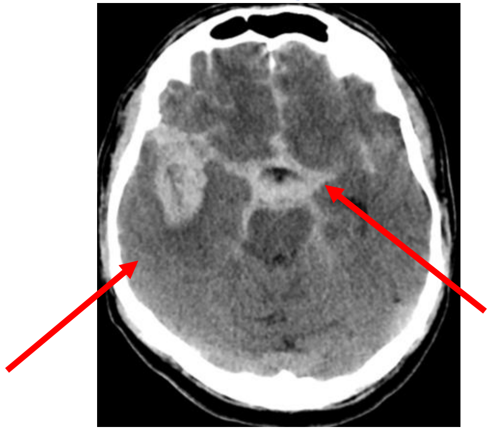
Which answer is <u>wrong</u> ?

A) Explosive, severe headache is a typical presentation of SAH
B) SAH are sometimes preceded by « sentinal headaches »
C) The most frequent cause of SAH is a ruptured saccular aneurysms of the circle of Willis
D) Acute non-contrast CT is often normal in SAH
E) Intracranial arterial vasospasm is a typical complication of SAH

# Subarachnoid hemorrhage

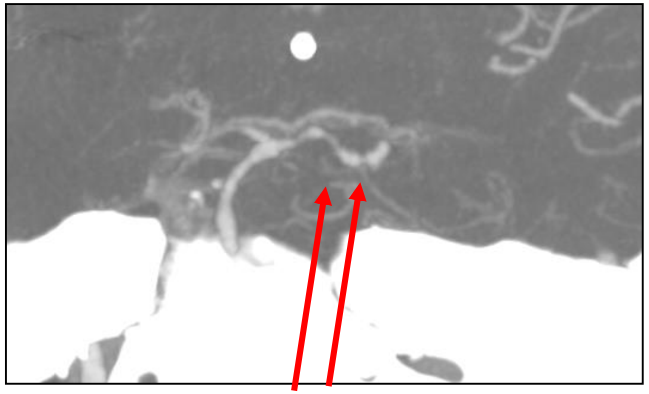
## from a ruptured saccular anerysm

Sentinel headache → Explosive HA



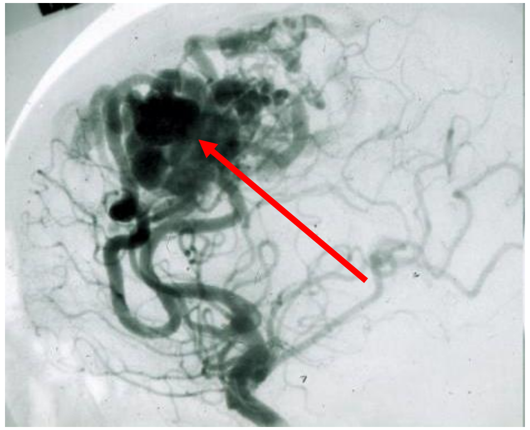
CT : subarachnoid blood
nearly always visible

CT-angio: saccular aneurysm



Complication: vasospasmes

# Vignette 6



A 58 year old man with acute aphasia is diagnosed with a left intracerebral haematoma, attributed to an arteriovenous malformation (AVM).

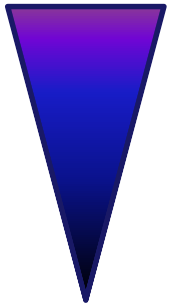
Which of the following is true ?

A) The most frequent presentation of AVM are epileptic seizures
B) Bleeding from AVMs is usually located in the deep brain structures
C) Digital substraction angiography (DSA) is the preferred method to characterize the AVM and to plan treatment
D) The annual bleeding risk in **a**symptomatic AVMs is 15-25%

# Malformations artérioveineuses (MAV)

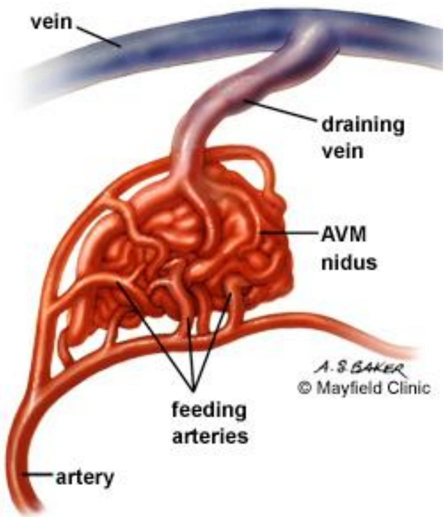
- Shunt artériel → veine de drainage

- Présentation clinique    Fréquence
  - Hémorragie
  - Epilepsie
  - Asymptomatiques
  - Autres



- Pronostic:
  - Spetzler-Martin grading system

- Risque annuel d'hémorragie env. 2 % pour les MAV **<u>a</u>**symptomatiques




# MAV : Diagnostic

## Digital subtraction angiography (DSA)

= Examen artériel le plus détaillé pour une suspicion de MAV

MAV dans le territoire de l'artère cérébrale antérieure droite

# Vignette 7b

A 58 year old man's AVM has had a recent rebleed.

He may be treated with any of the following modalities except :

A) Endovascular obliteration
B) Surgical resection of the hematoma and AVM
C) Prothrombotic drugs to prevent rebleeding
D) Radiosurgery
E) Abstention (i.e. no intervention)

# Vignette 8

A 58 year old hemiplegic woman arrives rapidly after an acute intracerebral bleed at the hospital. Which of the following treatment has been shown to improve her clinical outcome ?

A) Rapid haemostasis with tranexamic acid or recombinant factor VII

B) Mannitol to decrease intracranial pressure

C) Rapid and aggressive lowering of blood pressure to below 140/90

D) Evacuation of the blood by its repeated aspiration through a cranial burhole (craniopuncture)

# Vignette 8

A 74 year old woman is on oral anticoagulation for atrial fibrillation.

Each of the following measures decreases the risk of intracranial haemorrhages except ...

A) Good control of arterial hypertension
B) Avoidance of high INR on vitamine-K antagonists
C) Late start of anticoagulation after an ischemic stroke
D) Use of vitamine-K antagonists (rather than new oral anticoagulants)
E) Concomitant use of anticoagulants and antiplatelets

# Prévention de l'HIC

## chez les patients sous antiplaquettaires et anticoagulants

- Bien contrôler l'hypertension artérielle
- Eviter antiplaquettaires agressifs chez  patients cérébrovasculaires
- FA: utiliser les anticoagulants oraux directs (ACOD) à la place des AVK
- Eviter anticoagulation précoce après AVC ischémique
- Anti-Vitamines K : respecter fenêtre INR

ESO Cerebrovasc Dis 2008; MATCH Lancet 2004; SPS-3 NEJM 2012